Quick Quiz - Answer Key

For the parent: compare each answer and tick or cross

1. Simplify: $x^5 \times x^{-2}$

Multiply -> add exponents	$x^{5+(-2)}$
Answer:	**x^3**

2. Simplify: $(3a^2)^3$

Power goes to the 3 and to a^2	$3^3 \times a^{2 \times 3}$
Answer:	**$27a^6$**

3. Find the value (as a fraction): 2^{-4}

Negative -> move to bottom	$\frac{1}{2^4} = \frac{1}{16}$
Answer:	**1/16**

4. Find the value (as a fraction): $\left(\frac{2}{5}\right)^{-2}$

Negative exponent flips the fraction	$\left(\frac{5}{2}\right)^2 = \frac{25}{4}$
Answer:	**25/4**

5. Simplify (positive exponents only): $\frac{12x^5y^{-2}}{4x^2y^3}$

Numbers: 12 / 4 = 3	
x: 5 - 2 = 3	x^3
y: -2 - 3 = -5	$y^{-5} = \frac{1}{y^5}$
Answer:	**$3x^3 / y^5$**

6. Find the value: $-2^0 - (-2)^0$

$-2^0 = -(1) = -1$; $(-2)^0 = 1$	$-1 - 1$
Answer:	**-2**

7. Write in scientific notation: 0.00058

Move point 4 places right -> small number	5.8×10^{-4}
Answer:	**5.8×10^{-4}**

8. Write as an ordinary number: 4.07×10^5

Positive exponent -> move point 5 places right	
Answer:	**407,000**

9. Calculate. Give the answer in scientific notation: $(6 \times 10^4)(5 \times 10^3)$

Multiply numbers, add exponents	30×10^7
30 is too big -> 3.0, add 1 to exponent	3×10^8
Answer:	**3×10^8**

10. Calculate. Give the answer in scientific notation: $\frac{8 \times 10^9}{2 \times 10^4}$

Divide numbers, subtract exponents	$4 \times 10^{9-4}$
Answer:	**4×10^5**

11. Is each number rational or irrational? $\sqrt{49}$ $\sqrt{10}$

$\sqrt{49} = 7$ (a whole number)	
10 is not a perfect square	
Answer:	**$\sqrt{49}$ is rational ; $\sqrt{10}$ is irrational**

12. Between which two whole numbers is $\sqrt{50}$?

$49 < 50 < 64$	$\sqrt{49} = 7,\ \sqrt{64} = 8$
Answer:	**between 7 and 8**

13. Is -4 a whole number? Name all the sets it belongs to.

Whole numbers start at 0 (no negatives)	
Answer:	**No. -4 is an integer, rational and real**

14. List all the perfect squares between 50 and 150.

$8^2 = 64$, $9^2 = 81$, $10^2 = 100$, $11^2 = 121$, $12^2 = 144$	
Answer:	**64, 81, 100, 121, 144**

Score guide: 12-14 = ready | 9-11 = re-read the part with mistakes | below 9 = redo Part 1 and the quiz once more.